

Chong Tze Zhao

Proof over promise.

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EDUCATION

National University of Singapore

Singapore

Bachelor of Computing in Computer Science

August 2023 – May 2027 (Expected)

- NUS Merit Scholar; GPA: 4.58/5.00
- Relevant coursework: CS3233 Competitive Programming (advanced algorithms, C++ optimization), CS2100 Computer Organisation (memory hierarchy, pipelining)

TECHNICAL SKILLS

Languages: Python, C++, Go, Java, TypeScript, SQL

ML / Data / Systems: LightGBM, PyTorch, pandas, numpy, numba, PyArrow/Parquet, asyncio/WebSockets, Docker, Linux, SLURM, AWS (ECS/Fargate, S3, Lambda, CloudFormation), React, Express, FastAPI

Algorithms: competitive programming, single-threaded C++ optimization

INDEPENDENT SYSTEMATIC TRADING RESEARCH

Self-directed systematic-trading research, Feb–Jun 2026. For technical writeups and supporting artifacts, see chongtzezhao.github.io.

Validation-first trading research stack | *Python, LightGBM, numba, PyArrow, SLURM*

Feb – Jun 2026

- Built Python research infrastructure for tick ingestion, market-microstructure features, walk-forward evaluation, slippage calibration, leakage audits, and paper/live execution support.
- Implemented AFML-style validation (CUSUM events, triple-barrier labels, purged/embargoed CV, sample weights) with lookahead checks treated as pipeline gates rather than manual review.
- Calibrated slippage from real fills and rejected strategy variants that did not survive honest transaction costs, leakage checks, and walk-forward breadth.

SOFTWARE ENGINEERING EXPERIENCE

Software Engineer Intern

Jan 2026 – Jun 2026

Synapse (Singapore's national HealthTech agency)

Singapore

- Designed and shipped **two internal monitoring dashboards** covering scheduled batch jobs and Oracle Service Bus integration traffic; deployed to staging on **AWS ECS Fargate** behind an internal ALB.
- Architected a reusable decoupled **producer/consumer pattern** (Python writer → **S3** → Node/Express + React SPA) provisioned via **CloudFormation**, cutting time-to-build for the follow-on dashboard.
- Authored 2,000+ lines across **React 19 / TypeScript, Express 5 / Python**, and CloudFormation IaC, reaching **~90%** test coverage.

Software Engineer Intern

Jun 2025 – Aug 2025

Cinch Asia

Singapore

- Launched two company-wide AI services in **Python** and **Docker**, reducing manual data processing by **30%**.
- Rewrote 8 legacy **AWS Lambda** scripts into consolidated **Golang microservices**, reducing cold-start latency by **75%** and cutting compute costs by **20%**.
- Implemented CI/CD and monitoring pipelines (**GitHub Actions**, Docker) to automate deployment and improve reliability.

Teaching Assistant | *CS2100 Computer Organisation*

Aug 2024 – May 2025

National University of Singapore

Singapore

- Selected as Teaching Assistant for CS2100; taught 100+ students weekly in computer architecture, pipelining, memory hierarchy, assembly, and low-level performance optimization.

SELECTED PROJECTS

Exchange Matching Engine & Market Simulator | *C++17, price-time priority*

Jun 2026

- Built five layered implementations (reference oracle, intentionally broken, slow, cache-optimized, lock-free concurrent reads) sharing a deterministic workload; a full-trade-stream checksum oracle confirms bit-identical execution across correct engines and catches the deliberate mis-pricing bug.
- Optimized engine: flat price ladder, intrusive node pool, O(1) cancel; **CONSTRAINTS.md** documents exact breakdown conditions (sparse book, large id space, geometric resize spikes) rather than claiming universal wins.
- Parallel engine: seqlock for lock-free concurrent snapshot reads; validated under ASan, UBSan, and TSan against a sequential read oracle.

ACTIVITIES

Quant Engineer (early exposure)

Sep 2023 – May 2024

NUS Investment Society — Quantitative Finance Department

Singapore

- First exposure to systematic trading: implemented a **mean–CVaR** portfolio optimizer and backtested allocation strategies on historical market data for the department's research track.